



FT94X HTST/UHT Mini Pilot System

The Armfield FT94X HTST/UHT Mini Pilot System extends the range of continuous operation scaled-down units offered by Armfield to 200 l/hr (30-100 l/hr for the standard FT94X). The unit combines full flexibility with comprehensive instrumentation demanded of research and development equipment. The FT94X is designed for operation as both a stand-alone unit and also for seamless integration into a full processing system with the capability of aseptic packaging of processed product.

PRODUCT CODE: FT94X

Categories: Armfield Beverages Equipment, Armfield Dairy, Armfield Educational Food Technology, Armfield Liquid Foods, Armfield Nutraceuticals, Armfield Pharmaceuticals, FT Series - Educational Food Technology, FT94X HTST/UHT Mini Pilot System, IFT Series - Industrial Food Technology

Tags: ArmSOFT , Aseptic Fill , Aseptic Processing , Centrifugal Pump , CIP , Dataworx , Feed Vessel , Flow Meter , FT94X , Homogenisation , HTST , Mini Pilot System , PLC , Preheat , Process Flow , Realtime Comparisons , Service Unit , SIP , Suction Feed , Touch Screen , Two Stage Cooling , UHT

DESCRIPTION

The FT94X service unit contains the services, controls and instrumentation required to operate the tubular and plate heat exchangers described below. The FT94X may be specified with either or both of the heat exchangers. Switching between the two heat exchangers takes a matter of seconds.

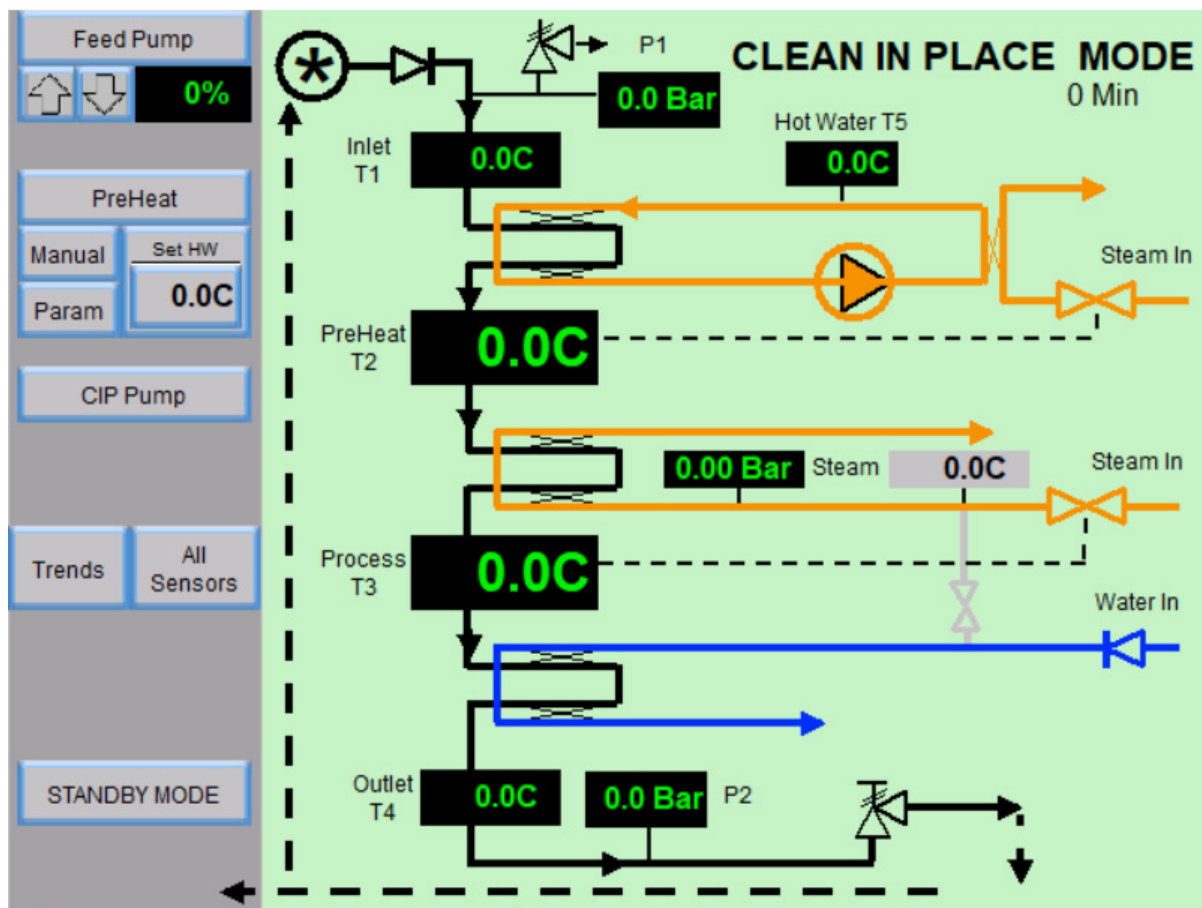
The service unit comprises: an eight inch touch screen control panel with the required controls and displays; a variable speed progressive cavity product feed pump with inlet level sensor and alarm to warn of low product level; a steam-heated hot water system for product preheating; automatic steam control valves for product preheating and final heating; a centrifugal CIP pump and other facilities required for CIP and SIP.

Sensors show product inlet, preheat, final heating and outlet temperatures. Product preheat and final heating temperatures are set by the operator. Inlet and outlet product pressure is also shown. There are various service sensors: pressure, level and temperature – see process flow diagram for details.

A sophisticated data logging package enables the measured data to be recorded onto a standard Windows PC (not supplied). The software records all temperatures, product and steam pressures, feed pump speed and many other senso.

values. Data from optional accessories (e.g. the flowmeter) is also recorded. Data may be displayed in tables and graphs (both real time and comparisons across previous runs), and may also be exported to MS Excel.

Interface: USB Software requirements: Win 98, ME, 2000, XP, Win 7/8/10



FT94X HTST/UHT Mini Pilot System Touch Screen

In addition FT94-59 Ethernet PLC update module is available. This accessory gives a direct link from a PC to update the software on the PLC using Dataworx™ Software (not provided).

TECHNICAL SPECIFICATIONS

FEED PUMP

Progressive cavity variable speed pump

Solids handling:	Soft 10 mm
Solids handling:	Hard 4 mm
Flow rate:	30-100 l/hr
Discharge pressure:	24 bar (max)

PRE-HEAT WATER SYSTEM

Water capacity:	~2-3 litres
Maximum temp:	110°C
Safety cut-outs:	Low level
Safety cut-outs:	Temperature
Safety cut-outs:	Product pump
Safety cut-outs:	Pressure relief valve

HEATING SYSTEM

External steam supply

Maximum temp:	165°C
Maximum temp:	Relief valve (7 barg)
Maximum temp:	Product pump off
Maximum temp:	steam valve closes

Max product heating duty: 17kW
Maximum temp: (preheat + heating)

TUBULAR HEAT EXCHANGER (FT94-20)

Number of tubes 10 (2 preheat, 4 heating, 4 cooling)

TUBE DIAMETER

(product side): 8.1 mm
Overall diameter: 22.0 mm
Length (heated): 1.46 m
Product volume: 1.0 l
Material: 316 stainless steel
Assembled test pressure: 30 bar
Working pressure: 24 bar (maximum)

PLATE HEAT EXCHANGER (FT94-30)

Plate overall dimensions: 150 x 480mm
Effective area per plate: 0.04m²
Plate thickness: 0.6mm
Product volume: 2.5litres
Plate Material: 316 stainless steel
Gasket Material: Food grade Viton
Max differential test pressure: 11.5bar
Max balanced test pressure: 12.5bar
Max working pressure: 9.6bar
Working temperature Max: 110°C
Min: -4°C

NUMBER OF PLATES

FT94-30 Preheating 13, Heating 6, Cooling 13
FT94-33 Secondary cooling 7

FEATURES & BENEFITS

- Compact design and easy installation
- Low product hold up
- Suitable for producing large batches of material for extensive product testing
- Suitable for a wide range of products due to both plate and tubular exchangers
- From cold start to steady state operation in seconds
- Rapid switchover from one heat exchanger to another
- Designed for easy integration into a process line
- Links directly to sterile filling bench or sealed bag filling systems
- Touch screen control for ease of use
- Compact, mobile, easy to install
- Continuous and batch operation
- Standard throughputs from 30 to >100 l/hr (dependent on conditions)
- Units available to 200 l/hr
- Process temperatures up to 150°C
- Capable of aseptic operation
- Fully instrumented
- Preheating by hot water (indirect)
- Heating by steam (indirect)
- Plate and tubular heat exchanger options
- Controlled product preheat and heating temperatures
- Simple and rapid switch over between heat exchangers



- Automatic product divert for low temperature on tubular heat exchanger
- Built in Clean In Place (CIP) and Sterilisation In Place (SIP) facilities
- Integral USB data logging
- Two-stage cooling option
- Electronic flowmeter option
- Suction feed with in-line level sensor

Applications

- Baby foods
- Beer
- Condiments
- Confectionery
- Culture media
- Desserts and puddings
- Fruit and vegetable purees
- Health and nutritional products
- Fruit juices and cordials
- Gelatine products
- Gravies
- Cream
- Ice cream
- Sauces and soups
- Yoghurts
- Meat pie fillings
- Milk
- Pet food
- Pharmaceuticals
- Ready meals

Options

TUBULAR HEAT EXCHANGER

The FT94-20 is a standard tubular heat exchanger design using a series of concentric 316 stainless steel tubes for product preheating, final heating and cooling. In all cases the product passes down the inner tube and the heating/cooling medium passes through the annulus.

Two heating tubes of 1.5 m length are used to preheat the product. Four similar tubes are used for product heating and an additional four tubes are used for product cooling.

Product is gently preheated using a hot water system, which is itself heated by steam via a secondary service heat exchanger. In the main heating section the product is indirectly heated using steam itself. Product is cooled normally using water or chilled water. The cooling section can be configured to enable two stages where the first stage is water/chilled water and the second stage is water/glycol.

The FT94-21 is a set of static mixers for two tubes. These significantly improve heat transfer with more viscous products and can be used for improving both heating and cooling efficiency. Multiple sets can be used to optimise both heating and cooling performance.

The FT94-22 product divert option enables product that has not been processed to a sufficiently high temperature to be diverted to drain thus ensuring product quality and heat treatment. The divert temperature is adjustable. Diversion is achieved using a low hold-up twin diaphragm valve block operated by compressed air and PLC controlled. Diverted product is cooled in a tubular heat exchanger before being led to drain.

The FT94-20 has a one second holding tube (at 50 l/hr) as standard. Contact Armfield for longer holding times.

PLATE HEAT EXCHANGER

The FT94-30 uses the sanitary heat exchanger employing the latest plate design for efficient heat transfer. Performance scale-up is straight forward. The standard unit is a three stage system with preheat (hot water) final heating (steam) and cooling (water/chilled water). The plates have a complex pressed profile to maximise turbulence and minimise dead zones. There is a relatively large gap between plates to enable handling of more viscous products. The plates are fitted with food grade Viton paraclip gaskets for easy replacement as these are simply clipped in place rather than glued. A number of options are available to enhance the performance of the standard plate heat exchanger. These options involve additional plate configurations, which can be supplied and fitted or may be easily retro-fitted by the user.

The FT94-32 homogeniser interconnecting plate enables the homogeniser to be connected downstream of the heating section. The plate is fitted into the cooling section.

The FT94-33 two-stage cooling option provides a complete additional cooling stage, which can be used in conjunction with a chilled water/glycol supply for the best possible product cooling. Output temperatures below 5°C are possible depending on the product and the flow rate. An interconnecting link is provided so that the two stages can be operated as a single stage when required.

The FT94-30 has a five second holding tube (at 50 l/hr) as standard. Contact Armfield for longer holding times.

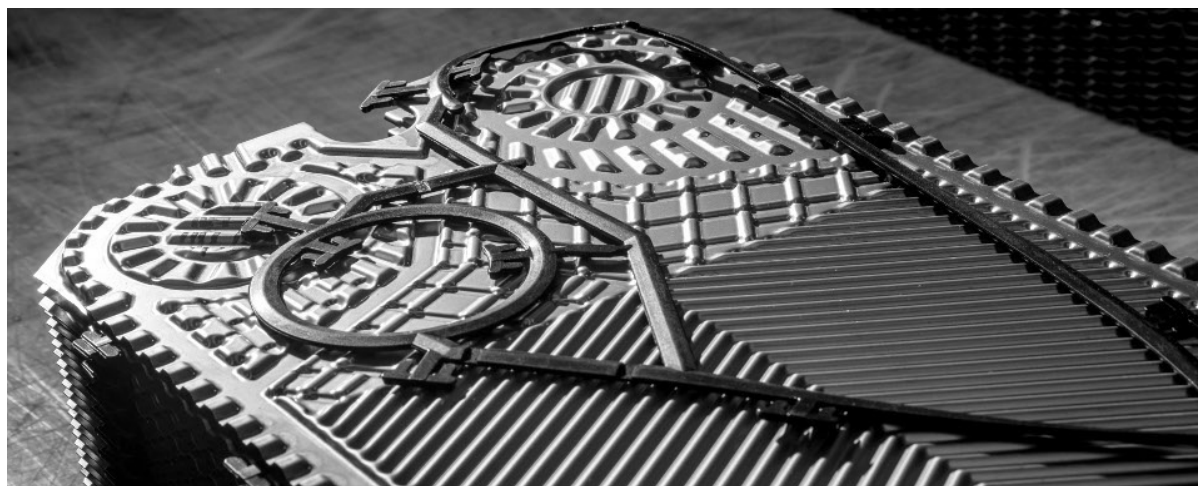


Plate Heat Exchanger Option

(FT94-33) TWO STAGE COOLING OPTION

FT94-33 Two stage cooling option for use with FT94-30 (PHE) to give much increased cooling capacity

FLOW METER OPTION (FT94-40)

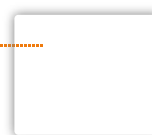
Type: Electromagnetic

Flow range: 30-100 l/hr

Accuracy: <1% error at 50 l/hr

SUCTION FEED

Suction feed with in-line float switch standard on the FT94X



FEED VESSEL (FT94-51)

The FT94-51 Feed Vessel option, is a 50 litre stainless steel vessel with lid, fitted with a product outlet valve and a drain (or recirculation) valve. The assembly is mounted on castors for ease of movement.



Armfield – Feed Vessel (FT94-51) Option

(FT94-59) ETHERNET PLC UPDATE MODULE

Comprising of a single port (RJ45) 10/100 ethernet connection module directly connected to the PLC. This module enables remote connection to the PLC for maintaining the most up to date software.

(FT94-87) STERILISATION LINK ACCESSORY

FT94-87 – Sterilisation Link Accessory: this option links the FT94X to a product storage tank. It enables pressurised hot water sterilisation of the FT94X and steam sterilisation of the product vessel so that the full system undergoes a monitored SIP procedure prior to sterile product filling to the vessel. There is a divert system so that interface material can be led to drain prior to switching to vessel filling.

Associated equipment



FT83-94 STERILE FILLER

- The FT83-94 enables sterile product produced by the FT94X to be packaged into pre-sterilised containers in a sterile environment – see separate data sheet
- The FT94X contains all the necessary functionality to sterilise the FT83-94
- The FT94 can also be interfaced directly to a sterile sealed bag filling system. **Contact Armfield for details.**
- Homogeniser subsystems
- Contact Armfield for details of the in-line Homogenisers for use with the FT94X.

Requirements

- **Electricity supply:** Single phase (see ordering codes)
- **Steam supply:** Heat output of 30kW (min 6.0 bar) – heating medium
- **Compressed air:** 7 l/s (min 7.0 bar, max 10.0 bar) – valve operation
- **Cooling water:** 10 l/min at 2 bar – product cooling
- **Chilled water:** Required if two-stage cooling option is specified

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 5.0m³

Gross weight: 708K

Dimensions

Length: 2.02m

Width: 0.70m

Height: 1.55m (Fitted with FT94-20)

Order Codes

- **FT94-A:** 220-240V/1ph/50Hz (10 A)
- **FT94-G:** 220V/1ph/60Hz (10 A)

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